Section 21.13 Optional Coils

Contents

1. INTRODUCTION

This chapter describes what actions should be taken when optional coils are installed on a Gyroscan Intera system. When adjustments are necessary, an installation document is provided together with the delivery of the coil. All optional coils are factory adjusted.
For corrective actions concerning coils, refer to the DSS program.

The next table gives an overview of coils and documents. The documents are present on the BBS in WP 5.1 format or word 6.0 and can be downloaded if necessary (Section MRI Download Area 9: Documentation). They can also be ordered via spare parts.

Coil Type	Document	
Quad T/L coil T5, T5-NT and Intera 0.5T	Service manual Quad T/L 21 Mhz	4522 983 54032
Quad Neck coil T5, T5-NT and Intera 0.5T	Service manual Quad Neck coil 21 Mhz	4522 983 54013
Breast coil T5, T5-NT and Intera 0.5T	Service manual Breast coil 21 Mhz	4522 983 54051
Endo Cavitary coil T5, T5-NT and Intera 0 5T	Service manual Endo coil 21 Mhz	4522 983 54071
ECCA Small	Service manual ECCA small	4522 981 07091
Endo coil S15-ACS, HQ, ACS-NT and Intera 1.5T	Service manual Endo coil 64 Mhz	4522 983 54081
Syn spine coil (all fieldstrengths)	Service manual Synergy spine coil	4522 981 04821
Syn. body coil ACS-NT, T10-NT and Intera 1.5T/1.0T	Service manual Synergy body coil	4522 981 04851
Syn. Cardiac coil ACS-NT, T10-NT and Intera 1.5T/1.0T	Service manual Synergy cardiac coil	4522 981 03871

2. INSTALLATION OPTIONAL COILS

2.1. SOFTWARE ADAPTIONS

To make new coils known to the system the new coil must be enabled in Gyrotool. Furthermore, the Preset Procedure protocols have to be filtered to add/adapt protocols for the new coil. The procedure for filtering of the preset protocols can be found in the Software Installation Manual Section: Adding new coils to the system.

2.1.1. ENABLING NEW COILS IN GYROTOOL

Login: GYROTOOL
Select: Modify System Configuration
Select: Coil administration
Select NT for each coil available on the system except for the next coils which are standard:
Intera 0.5T: Quad head coil
Quad body coil
Quad knee coil
Quad neck coil

Intera 1.5T OMNI: Quad body coil

2.2. HARDWARE ADAPTATIONS

Except for synergy coils, no extra hardware is necessary to use new coils. The procedure to install the synergy hardware is described in a separate manual. This is delivered with the optional synergy coils.

Optional coils delivered for Intera 1.5T and Intera 1.0T systems do not have to be adjusted during installation.

Some optional coils for Intra 0.5T systems have to be checked/adjusted to the correct frequency. The procedure for this adjustment is written in a Service Document delivered together with the coil. (see table)

2.3. FREQUENCY RECOGNITION

The coils for the different frequencies can be recognized with a coloured label on the cable:

Intera 0.5T:	21 MHz	blue label
Intera 1.0T:	42 MHz	yellow label
Intera 1.5T:	64 MHz	red label

2.4. CABLE ADAPTATION

The PICU (Patient Interface Control Unit) can be positioned on the left or on the right side of the system. The head coil and the quad neck coil need to have the cable on the same side as the PICU.
If this is not the case, the cable has to be adapted to the other side of the coil. For the head coil the cable adaptation procedure is written in the chapter RF-coils of this manual. For the quad neck coil 1.0T the cable adaptation procedure is printed on the inside of the coil. It becomes visible, when the bottom cover of the coil is removed. For the quad neck 0.5T and 1.5T the cable adaptation must be done as follows:

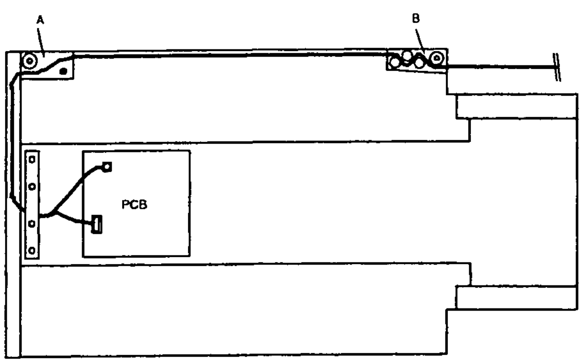

Figure 1 Cable routing

- remove the bottom cover
- remove also the small bottom cover so that the PCB is visible
- remove the rear cover with 5 screws
- remove the cable by disconnecting it from the PCB.
- reinstall the cable via the other side of the coil (holes are already present)
- remove the metal brackets A and B, and guide the cable through the cable-holders as indicated in Figure 1
- mount the metal brackets A and B.
- mount the covers again and take care that the covers are free from the surface of the patients table and that the coil is easy to slide.

3. CABLE TRAPS

Cable traps are used in combination with RF-coils.
The cable traps are frequency dependent. The different frequencies can be recognized by the coloured label on the cable·

Intera 0.5T	21 MHz	Blue label
Intera 1.0T	42 MHz	Yellow label
Intera 1.5T	64 MHz	Red label

WARNING

Always use a cable trap on a system with the corresponding frequency.
Using a cable trap on a system with another frequency may lead to dangerous situations.

4. IMAGE QUALITY

4.1. IMAGE QUALITY TEST

For every coil an Image Quality Test (IQT) procedure must be performed to check the correct functioning of the coil. When the SPT program is started, the coil can be selected and all necessary information to perform the Image Quality Test is presented on the screen. (see also chapter Acceptance of this SMI).

The procedure to start the program:

- Select : Scan control.
- Select : SPT scanning.
- Select : Enter service mode.
- Select : Perform IQP/EST measurements.
- Select : Analysis principle. Default is IQP.
- Select : Coil type

When the scans have finished, the images are evaluated automatically and an MRL printout can be made.

- Select : ICON "system performance tools".
- Select : Enter service mode
- Select : System verification
- Select : Measurement result lists

A list of available data_files is given:
SPT_* (in which the * stands for the examination_name)

- Select : the type of the MRL
- Select : sections analysis to be listed
- Select : output device(s)

4.2. IMAGE QUALITY SPECIFICATIONS

NOTE

The specifications for the Image Quality Test are present in the software except for new introduced optional coils. When new optional coils are introduced in this software release, the specifications can be found on the BBS.

The MRL output will show the specifications for image quality measurements. For new introduced optional coils, the specifications are not present in the software but must be downloaded from the BBS.

The image quality specifications are distributed via "spec-files". Two spec-files are used. One for the Standard coils and one for the NT coils. The spec-files can be found on the BBS in the "download files" area of the product group MR and on the DSS-II CD-ROM subdirectory D:\VARIOUS.MRI. The spec-files must be loaded into the GYRO$SPT_SPEC directory and renamed using the following naming convention.

Spec_file: aa**R**b**V**c**L**d_ee_**L**ff_gg.**SPEC**
aa : T5, T10, or ACS
b : Release nr.
c : Version nr.
d : Level nr.
ee : PowerTrak configuration (00 = PT1000, 01 = PT1000 Advanced, 02 = PT3000, 03 = PT6000).
ff : Level spec-file.
gg : ST or NT coils.

Example: the file name for R6.2.1 PowerTrak 6000, NT Coils, first update = ACSR6V2L1_03_L01_NT.SPEC

Please refer to chapter "Operator console" paragraph "connecting a service PC" to the host and "copying files from/to the NT system to/from the service PC" for more information about how to transfer the files.

5. Q IS MEASURED 1

During performance measurements of optional coils, sometimes the value "1" is found for the Q-factor. This is the result of the shift of the coil due to the presence of a phantom The Q-curve can not be measured completely with "1" as result. This is not a malfunction of the system or coil. The Q-factor is not specified any more for coils that experience this phenomena. For the latest specifications, the BBS spec file should be downloaded.

6. CORRECTIVE ACTIONS

In case the measured values for the Image Quality are not within specification check that the system is working within specification by performing image quality checks for head and/or body coil. Also check that the phantoms are correct e.g. with use of flexible surface coils.

In case the cable of the optional coil is damaged: in most cases the cable can be exchanged.

When the optional coil itself is suspected, please refer to DSS for further corrective actions.

6.1. COIL ID GIVES -1

In NT systems, the ID code of a coil can be found via a tool in the test software (see paragraph 7.1). When the connected coil is defective, the correct ID code can not be determined. The system will respond with the code "-1".
Check if other coils can be recognized to verify the correct working of the system.
Use the fault find procedures in DSS to repair the problem.

6.2. PRESET PROCEDURES MISSING FOR A COIL

After installation of the coil in the coil administration of the system (Gyrotool), a filtering of the protocols must be done according the software installation manual.
When for a specific coil, no preset protocols are present, check that the setting in the coil administration is correct.
When the setting was incorrect, change the setting to the correct value. The filtering of the preset protocols have to be repeated to make the protocols available for this coil Refer to the Release Bulletin for filtering procedure.